

Trade with Increasing Returns and Heterogeneous Firms

A fully derived study companion: Krugman, Melitz, and Melitz–Pareto
with lesson quizzes and a final exam

Study notes

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Part I

Monopolistic Competition and Increasing Returns (Krugman)

In neoclassical trade models (Ricardo, Heckscher–Ohlin, DFS, EK), trade is driven by *comparative advantage*: countries differ, and differences create gains from exchange. Krugman’s revolutionary point: even *identical* countries gain from trading with each other, because of **increasing returns to scale** and **love of variety**. This part builds that argument twice — first in the flexible Krugman-1979 version, then in the tractable CES Krugman-1980 version that plugs into gravity.

1 Lesson 1: The closed economy with variable markups (Krugman 1979)

1.1 Setup

Definition: Monopolistic competition

A market structure with two halves: *monopolistic* — each firm is the sole producer of its own differentiated variety, so it faces a downward-sloping demand curve and charges a markup over marginal cost; *competitive* — entry is free, so profits are driven to zero in equilibrium. The number (mass) of varieties is **endogenous**: it is determined by how many firms the market can sustain at zero profit.

Population is L ; labor is the only factor. Preferences of the representative household are additively separable over the set Ω of available varieties:

$$U = \int_{\omega \in \Omega} v(c(\omega)) d\omega, \quad v' > 0, v'' < 0.$$

The key behavioral assumption is on the **price elasticity of demand**,

$$\varepsilon(\omega) = -\frac{\partial \ln c(\omega)}{\partial \ln p(\omega)} > 0, \quad \frac{\partial \varepsilon(\omega)}{\partial c(\omega)} < 0 :$$

demand becomes *less* elastic the more of a variety people already consume.

Intuition: Why assume ε falls with c ?

Think of the first unit of a new good versus the tenth: when you consume little of something, small price changes easily push you toward substitutes (high elasticity); when it is a staple, you are less price-sensitive. This single assumption is what makes markups *respond to market size* in Krugman-1979 — and it is exactly what CES preferences will shut down in Krugman-1980, where ε is constant by construction.

On the supply side, a firm pays a fixed cost f (in labor) to operate, and then produces with constant labor productivity φ :

$$l(\omega) = f + \frac{1}{\varphi} q(\omega) \implies \pi(\omega) = \underbrace{p(\omega)q(\omega) - \frac{w q(\omega)}{\varphi}}_{\text{variable profits}} - \underbrace{wf}_{\text{fixed cost}}.$$

Intuition: Where are the increasing returns?

Average cost is $w(f/q + 1/\varphi)$, which *falls* with output: the fixed cost is spread over more units. That is internal economies of scale — the engine of the whole model. It also explains why each variety is made by exactly one firm: two firms splitting a variety would duplicate f pointlessly.

All firms share the same f and φ , so equilibrium is symmetric ($p(\omega) = p$, $c(\omega) = c$) and we can drop ω . Market clearing: goods, $q = Lc$; labor, $L = M[f + q/\varphi]$ where $M \equiv |\Omega|$ is the number of firms/varieties.

1.2 The two equilibrium curves

Derivation: The pricing curve (PP)

The firm chooses p to maximize $pLc(p) - wf - \frac{w}{\varphi}Lc(p)$, taking aggregate variables as given. Differentiate with respect to p (the L 's divide out):

$$c + p \frac{\partial c}{\partial p} - \frac{w}{\varphi} \frac{\partial c}{\partial p} = 0.$$

Divide by c and use $\varepsilon = -\frac{\partial c}{\partial p} \frac{p}{c}$:

$$1 - \varepsilon + \frac{w}{\varphi} \frac{\varepsilon}{p} = 0 \iff p \left(1 - \frac{1}{\varepsilon}\right) = \frac{w}{\varphi} \iff \boxed{\frac{p}{w} = \frac{1}{\varphi} \frac{\varepsilon}{\varepsilon - 1}} \quad (\text{PP})$$

This is marginal revenue = marginal cost. Each firm charges markup $\mu = \frac{\varepsilon}{\varepsilon - 1}$ over marginal cost w/φ .

Slope: since ε is *decreasing* in c , and $\mu = \varepsilon/(\varepsilon - 1)$ is decreasing in ε , the markup is *increasing* in c . So PP slopes **up** in the $(c, p/w)$ plane: more consumption per head $\rightarrow$ less elastic demand $\rightarrow$ fatter markup.

Derivation: The free-entry curve (ZZ)

Free entry pushes profits to zero:

$$\pi = pq - w \left[f + \frac{q}{\varphi} \right] = 0 \implies p = w \frac{f + q/\varphi}{q},$$

i.e. price equals *average* cost (the Chamberlinian tangency). With $q = Lc$:

$$\boxed{\frac{p}{w} = \frac{f + Lc/\varphi}{Lc} = \frac{1}{\varphi} + \frac{f}{Lc}} \quad (\text{ZZ})$$

Slope: p/w is *decreasing* in c — when each person buys more, the fixed cost is spread over more units, so a lower price is compatible with zero profit. ZZ slopes **down**.

Derivation: Solving the closed economy

PP (up-sloping) and ZZ (down-sloping) intersect once, pinning down $(c_0, (p/w)_0)$. Setting $PP = ZZ$:

$$\frac{1}{\varphi} \frac{\varepsilon}{\varepsilon - 1} = \frac{1}{\varphi} + \frac{f}{Lc} \implies \frac{f}{Lc} = \frac{1}{\varphi} \left(\frac{\varepsilon}{\varepsilon - 1} - 1 \right) = \frac{1}{\varphi(\varepsilon - 1)} \implies \boxed{q = Lc = (\varepsilon - 1) \varphi f}$$

Then labor market clearing delivers the number of varieties:

$$M = \frac{L}{f + \frac{1}{\varphi} Lc} = \frac{L}{f + (\varepsilon - 1)f} = \boxed{\frac{L}{\varepsilon f}}$$

Intuition: Reading the solution

Firm size $q = (\varepsilon - 1)\varphi f$ is pinned down by the tension between scale economies (f pushes firms to be big) and market power (ε close to 1 lets firms survive while small). The variety count $M = L/(\varepsilon f)$ says: a bigger economy supports *more firms*, not (in this closed equilibrium) bigger ones; cheaper entry (lower f) and less elastic demand also raise variety.

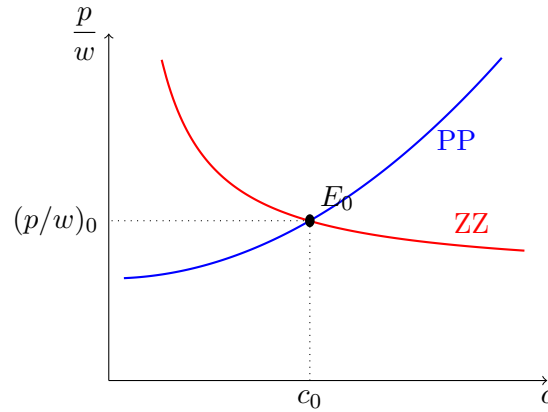


Figure 1: Closed-economy equilibrium (Krugman 1979). The upward-sloping pricing curve PP and the downward-sloping free-entry curve ZZ intersect once at E_0 , determining per-capita consumption c_0 and the price-wage ratio $(p/w)_0$.

Understanding Quiz 1 — Lesson 1

Q1.1 In Krugman-1979 the markup $\mu = \varepsilon/(\varepsilon - 1)$ rises when per-capita consumption c rises. The chain of logic is:

- (A) $c \uparrow \Rightarrow$ demand more elastic $\Rightarrow$ markup falls
- (B) $c \uparrow \Rightarrow$ demand less elastic $\Rightarrow$ markup rises
- (C) $c \uparrow \Rightarrow$ marginal cost falls $\Rightarrow$ markup rises
- (D) $c \uparrow \Rightarrow$ fixed costs are spread $\Rightarrow$ markup rises

Q1.2 The ZZ curve slopes down in $(c, p/w)$ space because:

- (A) higher consumption makes demand more elastic

- (B) firms with higher output spread the fixed cost over more units, so zero profit requires a lower price
- (C) the wage falls when consumption rises
- (D) entry costs rise with consumption

Q1.3 If the fixed cost f doubles (all else fixed, including ε at equilibrium), the number of varieties $M = L/(\varepsilon f)$:

- (A) doubles
- (B) halves
- (C) is unchanged
- (D) rises by a factor ε

2 Lesson 2: Opening to trade — pro-competitive gains and variety gains

Now let two *identical* such economies open to costless trade. Since countries are clones, there is no comparative advantage whatsoever — and yet there will be gains.

Derivation: What changes: the market doubles, the workforce does not

The free-trade equilibrium system is the same as before except that each firm can now sell to $2L$ consumers:

$$\frac{p}{w} = \frac{1}{\varphi} \frac{\varepsilon}{\varepsilon - 1} \quad (\text{PP, unchanged}), \quad \frac{p}{w} = \frac{1}{\varphi} + \frac{f}{2Lc} \quad (\text{ZZ, with } q = 2Lc),$$

$$M = \frac{L}{f + q/\varphi} \quad (\text{Labor MCC, unchanged } L).$$

At every level of c , the new ZZ lies *below* the old one (the $f/(2Lc)$ term is smaller): **ZZ shifts down, PP does not move**. The new intersection E_1 therefore has

$$\left(\frac{p}{w}\right)_1 < \left(\frac{p}{w}\right)_0 \quad \text{and} \quad c_1 < c_0.$$

Derivation: Firm size rises, domestic variety falls, total variety rises

Firm size. Goods market clearing alone ($q = 2Lc$, with c lower) is ambiguous. Instead rewrite free entry:

$$pq = w \left[f + \frac{q}{\varphi} \right] \implies q \left(\frac{p}{w} - \frac{1}{\varphi} \right) = f \implies q = \frac{f}{p/w - 1/\varphi}.$$

Since $(p/w)_1 < (p/w)_0$, the denominator shrinks, so $\boxed{q_1 > q_0}$: surviving firms are *bigger*, exploiting scale economies.

Varieties produced per country. From labor market clearing $M = L/[f + q/\varphi]$ with $q_1 > q_0$: $\boxed{M_1 < M_0}$ — fewer, larger firms at home.

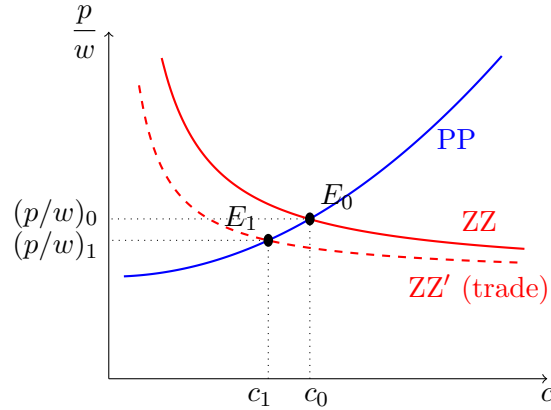


Figure 2: Free-trade equilibrium (Krugman 1979). Doubling market size shifts the free-entry curve down to ZZ' while PP is unchanged: the new equilibrium E_1 has a lower price–wage ratio (pro-competitive effect) and lower per-capita consumption of each variety.

Varieties consumed. Consumers now access both countries' menus, $2M_1$ varieties. Is $2M_1 > M_0$? Compare

$$2M_1 = \frac{2L}{f + \frac{1}{\varphi} 2Lc_1} \quad \text{vs} \quad M_0 = \frac{L}{f + \frac{1}{\varphi} Lc_0}.$$

Cross-multiplying, $2M_1 > M_0$ iff $2(f + \frac{1}{\varphi} Lc_0) > f + \frac{1}{\varphi} 2Lc_1$, i.e. iff $f + \frac{2L}{\varphi}(c_0 - c_1) > 0$ — true, since $c_0 > c_1$. So $\boxed{2M_1 > M_0}$.

Intuition: Two brand-new sources of gains from trade

Neither channel exists in neoclassical models: (a) *The pro-competitive effect.* With a bigger market, each consumer buys less of each variety (c falls), demand becomes *more* elastic, and markups get squeezed: (p/w) falls, so the real wage w/p rises. Trade disciplines market power. (b) *The variety effect.* Consumers value variety ($v'' < 0$ means marginal utility of any one good falls fast — spreading consumption over more goods is better). Trade shrinks each country's menu but more than doubles the accessible menu: $2M_1 > M_0$. And note the reallocation flavor already present: fewer, bigger firms — trade rationalizes the industry. Melitz will make this rationalization *selective* once firms differ.

Understanding Quiz 2 — Lesson 2

Q2.1 When two identical Krugman-1979 economies open to trade, which curve moves in the $(c, p/w)$ diagram?

- (A) PP shifts up; ZZ fixed
- (B) PP shifts down; ZZ fixed
- (C) ZZ shifts down; PP fixed
- (D) Both shift up

Q2.2 After opening to trade, output per surviving firm q rises because:

- (A) consumers buy more of each variety
- (B) zero profits with a lower price-to-wage ratio require spreading the fixed cost over more units
- (C) the labor force doubles
- (D) productivity φ increases

Q2.3 Which statement about varieties is correct after trade opening?

- (A) Produced per country: up; available to consumers: up
- (B) Produced per country: down; available to consumers: down
- (C) Produced per country: down; available to consumers: up
- (D) Produced per country: up; available to consumers: down

3 Lesson 3: Krugman 1980 — CES, gravity, and the home-market effect

Krugman-1980 trades generality for tractability: preferences become CES, so demand elasticity — and hence the markup — is *constant*. The payoff: a clean multi-country gravity model with endogenous entry.

3.1 Demand and pricing

N countries, labor endowments L_i , iceberg costs $d_{ij} \geq 1$. Preferences in country j are CES with elasticity $\sigma > 1$ over all varieties from all origins:

$$U_j = \left[\sum_{i=1}^N \int_{\omega \in \Omega_i} c_{ij}(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right]^{\frac{\sigma}{\sigma-1}} \implies x_{ij}(\omega) \equiv p_{ij}(\omega) c_{ij}(\omega) = \left(\frac{p_{ij}(\omega)}{P_j} \right)^{1-\sigma} X_j,$$

with X_j total expenditure and $P_j = \left[\sum_i \int_{\Omega_i} p_{ij}(\omega)^{1-\sigma} d\omega \right]^{\frac{1}{1-\sigma}}$ the Dixit–Stiglitz price index.

Derivation: Constant markup pricing

A firm in i delivering to market j faces demand with constant elasticity σ and marginal delivered cost $d_{ij}w_i/\varphi_i$. Apply the Lesson-1 pricing rule with $\varepsilon = \sigma$ (constant):

$$p_{ij} = \frac{\sigma}{\sigma-1} \frac{d_{ij}w_i}{\varphi_i} \implies p_i(\omega) = p_i = \frac{\sigma}{\sigma-1} \frac{w_i}{\varphi_i} \text{ at the factory gate.}$$

Because σ never moves, **markups no longer respond to market size** — the pro-competitive channel of 1979 is switched off by assumption. What survives is scale economies + variety.

Derivation: Free entry pins down firm size and the mass of firms

Variable profit per unit is $p_i - \frac{w_i}{\varphi_i} = \frac{\sigma}{\sigma-1} \frac{w_i}{\varphi_i} - \frac{w_i}{\varphi_i} = \frac{1}{\sigma-1} \frac{w_i}{\varphi_i}$. Free entry equates total variable profits to fixed costs:

$$\frac{1}{\sigma-1} \frac{w_i}{\varphi_i} q_i = w_i f_i \implies \boxed{q_i = (\sigma-1) \varphi_i f_i}$$

where $q_i = \sum_j d_{ij} c_{ij}$ is total production (inclusive of iceberg “melt”). Labor market clearing:

$$L_i = M_i \left[f_i + \frac{q_i}{\varphi_i} \right] = M_i [f_i + (\sigma - 1)f_i] = M_i \sigma f_i \quad \implies \quad \boxed{M_i = \frac{L_i}{\sigma f_i}}$$

The mass of varieties is proportional to population — market size maps one-for-one into product diversity, not firm scale. (Compare Lesson 1: same structure with $\varepsilon = \sigma$.)

3.2 Gravity

Derivation: Trade flows and shares

Aggregate exports from i to j = (mass of firms) $\times$ (exports per firm):

$$X_{ij} = M_i \frac{(d_{ij} p_i)^{1-\sigma}}{P_j^{1-\sigma}} X_j, \quad P_j = \left[\sum_i M_i (d_{ij} p_i)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}, \quad X_j = w_j L_j.$$

Substituting $p_i = \frac{\sigma}{\sigma-1} \frac{w_i}{\varphi_i}$ and $M_i = L_i/(\sigma f_i)$, the trade share becomes

$$\boxed{\pi_{ij} = \frac{\bar{\sigma} \frac{L_i}{f_i} \varphi_i^{\sigma-1} (d_{ij} w_i)^{1-\sigma}}{\sum_\ell \bar{\sigma} \frac{L_\ell}{f_\ell} \varphi_\ell^{\sigma-1} (d_{\ell j} w_\ell)^{1-\sigma}}}$$

with $\bar{\sigma}$ a constant. General equilibrium: wages $\{w_i\}$ such that these shares and price indices hold and budgets balance, $w_i L_i = \sum_j \pi_{ij} w_j L_j$.

Intuition: One gravity equation, three different stories

Set the three workhorse models side by side:

$$\text{Krugman: } \pi_{ij} \propto \frac{L_i}{f_i} \varphi_i^{\sigma-1} (d_{ij} w_i)^{1-\sigma}, \quad \text{EK: } \pi_{ij} \propto T_i (d_{ij} w_i)^{-\theta}, \quad \text{Armington: } \pi_{ij} \propto b_i a_i^{\sigma-1} (d_{ij} w_i)^{1-\sigma}.$$

All are “origin shifter $\times$ (cost)^{−elasticity}”, normalized by destination competition.” They differ in what the shifter *means* (number of firms and productivity; technology stock; endowment/taste) and in what the elasticity *is* ($1 - \sigma$, a taste parameter, in Krugman and Armington; $-\theta$, a technology-heterogeneity parameter, in EK). Aggregate trade data alone cannot tell these stories apart — micro data can.

3.3 Two famous implications

Intuition: Home-market effect

Countries tend to *export* the goods for which they have large *domestic* demand. Why? With scale economies, production wants to concentrate in one place; with trade costs, the profit-maximizing place is near the big market — produce where demand is, export to the rest. This reverses neoclassical logic, where large demand for a good makes you an *importer* of it.

(Davis 1998 and Costinot et al. 2019 test this.)

Derivation: Population effect: bigger countries are richer

Welfare is real income $U_i = w_i/P_i$. Evaluate the *domestic* trade share:

$$\pi_{ii} = \bar{\sigma} L_i f_i^{-1} \varphi_i^{\sigma-1} d_{ii}^{1-\sigma} \frac{w_i^{1-\sigma}}{P_i^{1-\sigma}} = \bar{\sigma} L_i f_i^{-1} \varphi_i^{\sigma-1} d_{ii}^{1-\sigma} \left(\frac{w_i}{P_i} \right)^{1-\sigma}.$$

Solve for real income (raise to $\frac{1}{\sigma-1}$, noting the $1-\sigma$ exponent flips the share to the denominator):

$$\frac{w_i}{P_i} = \left[\frac{\bar{\sigma} L_i \varphi_i^{\sigma-1} d_{ii}^{1-\sigma}}{f_i \pi_{ii}} \right]^{\frac{1}{\sigma-1}} = A_i L_i^{\frac{1}{\sigma-1}} \pi_{ii}^{-\frac{1}{\sigma-1}}, \quad A_i \equiv (\bar{\sigma}/f_i)^{\frac{1}{\sigma-1}} \frac{\varphi_i}{d_{ii}}.$$

Two lessons: (i) *controlling for openness* (π_{ii}), more populous countries are richer, with elasticity $\frac{1}{\sigma-1} > 0$ — more people $\Rightarrow$ more varieties $\Rightarrow$ higher real income (a pure scale effect); (ii) openness itself raises welfare: lower π_{ii} raises w_i/P_i (the exponent on π_{ii} is negative), the same sufficient-statistic logic as in EK. In the data the scale effect is muted because large countries also have larger internal frictions d_{ii} and higher π_{ii} (Ramondo–Rodríguez-Clare–Saborío 2016).

Understanding Quiz 3 — Lesson 3

Q3.1 In Krugman-1980 markups are constant because:

- (A) firms collude on a fixed markup
- (B) CES preferences imply a constant price elasticity of demand σ
- (C) free entry fixes the markup at average cost
- (D) iceberg trade costs neutralize elasticity changes

Q3.2 The trade elasticity (of flows with respect to trade costs) in Krugman-1980 is:

- (A) $-\theta$, a technology parameter
- (B) $1-\sigma$, a taste/substitution parameter
- (C) -1 always
- (D) $\sigma\theta$

Q3.3 The home-market effect says that, with scale economies and trade costs, a country with unusually large demand for a good will tend to:

- (A) import that good, since local demand exhausts supply
- (B) export that good, since production concentrates near large markets
- (C) neither import nor export it
- (D) subsidize it

Part II

Heterogeneous Firms (Melitz)

4 Lesson 4: What firm-level data revealed

In Krugman's world all firms are clones: every firm exports, and exports the same amount. When firm-level data arrived in the 1990s (Bernard, Jensen, Redding, Schott 2007 is the classic summary), reality looked very different.

Definition: The stylized facts

Exporting is rare. In 2000 only about 4% of U.S. firms exported at all; in manufacturing (2002 Census) about 18% of firms exported, and even exporters shipped only ~14% of output abroad.

Exports are concentrated. The top 10% of exporters accounted for 96% of total U.S. exports; firms exporting 5+ products to 5+ countries were 11.9% of exporters but 92% of export value — strongly suggestive of a *fat-tailed* (Pareto-like) size distribution.

Exporters are different (“export premia”). Compared with non-exporters in the same industry, exporters are larger (log employment premium ≈ 0.97), more productive, higher-wage, more capital- and skill-intensive.

Both margins matter. Exporters make more products (extensive) *and* ship more per product (intensive).

Intuition: Why this kills the representative firm

If exporting were just a matter of shipping costs, firms would differ only in degree. Instead the data show *selection*: a small elite of exceptional firms does all the exporting. That pattern demands a model where (i) firms differ in productivity, and (ii) there is a *fixed cost of exporting* so only the best find it worthwhile. That model is Melitz (2003) — and it delivers a new gain from trade: **reallocation of resources toward more productive firms**, raising aggregate productivity.

Understanding Quiz 4 — Lesson 4

Q4.1 Roughly what fraction of U.S. firms were exporters in 2000?

- (A) 4%
- (B) 25%
- (C) 50%
- (D) 96%

Q4.2 “The top 10% of exporters account for 96% of exports” is most directly evidence of:

- (A) constant returns to scale
- (B) a fat-tailed firm size distribution
- (C) balanced trade
- (D) identical firms with random shipping shocks

Q4.3 Which is NOT one of the documented exporter premia?

- (A) higher employment

- (B) higher productivity
- (C) higher capital intensity
- (D) lower wages

5 Lesson 5: The Melitz model, closed economy

5.1 Setup: Krugman 1980 + heterogeneity (Hopenhayn 1992)

Firms share the operating fixed cost f but differ in productivity φ : labor requirement $l(\varphi) = f + q/\varphi$. A firm is indexed by its φ . Normalize the wage to $w = 1$.

Derivation: Firm-level prices, sales, profits

CES demand with elasticity σ and constant-markup pricing give

$$p(\varphi) = \frac{\sigma}{\sigma-1} \frac{1}{\varphi}, \quad q(\varphi) = \frac{R}{P} \left(\frac{p(\varphi)}{P} \right)^{-\sigma}, \quad r(\varphi) = p(\varphi)q(\varphi) = R \left(\frac{p(\varphi)}{P} \right)^{1-\sigma},$$

with $R = L$ aggregate income and P the price index. **Profit as a fixed share of revenue:** variable cost is $q/\varphi = r/(p\varphi) = r \frac{\sigma-1}{\sigma}$, so variable profit is $r - r \frac{\sigma-1}{\sigma} = r/\sigma$ and

$$\pi(\varphi) = \frac{r(\varphi)}{\sigma} - f$$

The key ratio. Since $p(\varphi) \propto 1/\varphi$,

$$\frac{r(\varphi_1)}{r(\varphi_2)} = \left(\frac{p(\varphi_1)}{p(\varphi_2)} \right)^{1-\sigma} = \left(\frac{\varphi_1}{\varphi_2} \right)^{\sigma-1}.$$

Relative revenues depend only on relative productivities — the workhorse identity of everything below.

Derivation: Aggregation: the representative firm $\tilde{\varphi}$

Let $\mu(\varphi)$ be the density of active firms and M their mass. The price index:

$$P^{1-\sigma} = \int_0^\infty p(\varphi)^{1-\sigma} M \mu(\varphi) d\varphi = M \left(\frac{\sigma}{\sigma-1} \right)^{1-\sigma} \int_0^\infty \varphi^{\sigma-1} \mu(\varphi) d\varphi.$$

Define the CES-weighted average productivity

$$\tilde{\varphi} = \left[\int_0^\infty \varphi^{\sigma-1} \mu(\varphi) d\varphi \right]^{\frac{1}{\sigma-1}} \implies P = M^{\frac{1}{1-\sigma}} \frac{\sigma}{\sigma-1} \frac{1}{\tilde{\varphi}} = M^{\frac{1}{1-\sigma}} p(\tilde{\varphi}).$$

The whole distribution collapses into one number: aggregates behave *as if* there were M identical firms with productivity $\tilde{\varphi}$: $R = Mr(\tilde{\varphi})$, $\Pi = M\pi(\tilde{\varphi})$. Note $\tilde{\varphi}$ is not the arithmetic mean — it weights by $\varphi^{\sigma-1}$, i.e. by market success.

5.2 Entry, exit, and the two equilibrium conditions

Definition: Entry technology and the cutoff

To enter, a prospective firm pays a sunk cost f_e and *then* draws φ from a known distribution (pdf g , cdf G) — entry is a lottery. A firm with $\pi(\varphi) < 0$ exits immediately; since π is increasing in φ , exit is a **cutoff rule**: stay iff $\varphi \geq \varphi^*$, where $\pi(\varphi^*) = 0$. Each period, incumbents die with probability δ , so a survivor's expected value is $\sum_{t=0}^{\infty} (1 - \delta)^t \pi(\varphi) = \pi(\varphi)/\delta$. The distribution of *active* firms is the entry distribution truncated at φ^* : $\mu(\varphi) = g(\varphi)/[1 - G(\varphi^*)]$ for $\varphi \geq \varphi^*$; hence $\tilde{\varphi} = \tilde{\varphi}(\varphi^*)$ is increasing in the cutoff.

Derivation: Zero Cutoff Profit (ZCP)

Average profit $\bar{\pi} = \pi(\tilde{\varphi})$. Use the revenue ratio: $r(\tilde{\varphi}) = (\tilde{\varphi}/\varphi^*)^{\sigma-1} r(\varphi^*)$. The cutoff firm breaks even: $\pi(\varphi^*) = 0 \Rightarrow r(\varphi^*) = \sigma f$. Therefore

$$\bar{\pi} = \frac{r(\tilde{\varphi})}{\sigma} - f = \left(\frac{\tilde{\varphi}(\varphi^*)}{\varphi^*} \right)^{\sigma-1} f - f = f k(\varphi^*), \quad k(\varphi^*) \equiv \left(\frac{\tilde{\varphi}(\varphi^*)}{\varphi^*} \right)^{\sigma-1} - 1. \quad (\text{ZCP})$$

$k(\varphi^*)$ can be shown to be decreasing in φ^* (a higher cutoff compresses the surviving distribution toward its bottom): **ZCP slopes down** in $(\varphi^*, \bar{\pi})$ space.

Derivation: Free Entry (FE)

An entrant pays f_e , succeeds (draws $\varphi \geq \varphi^*$) with probability $1 - G(\varphi^*)$, and a successful draw is worth $\bar{\pi}/\delta$ on average. Free entry means zero expected net value:

$$[1 - G(\varphi^*)] \frac{\bar{\pi}}{\delta} = f_e \quad \Longleftrightarrow \quad \bar{\pi} = \frac{\delta f_e}{1 - G(\varphi^*)}. \quad (\text{FE})$$

FE slopes up: a higher cutoff makes successful entry rarer, so the survivors' average profit must be higher to keep entrants indifferent.

The intersection of ZCP (down) and FE (up) pins down $(\varphi^*, \bar{\pi})$ uniquely. The mass of firms then follows from accounting: aggregate profits $\Pi = R/\sigma - Mf$ and $\Pi = M\bar{\pi}$, $R = M\bar{r}$ give $\bar{r} = \sigma(\bar{\pi} + f)$ and $M = R/\bar{r} = L/[\sigma(\bar{\pi} + f)]$.

Intuition: Closed economy: heterogeneity is invisible in aggregates

Strikingly, the closed-economy *aggregate* predictions are those of Krugman-1980 with a representative firm $\tilde{\varphi}$: welfare is $W = P^{-1} = M^{\frac{1}{1-\sigma}} \frac{\sigma-1}{\sigma} \tilde{\varphi}$. Heterogeneity earns its keep only when trade is costly *and* exporting has a fixed cost — then market access differs across firms, selection bites, and the distribution starts moving aggregates.

Understanding Quiz 5 — Lesson 5

Q5.1 In Melitz, firm profit is $\pi(\varphi) = r(\varphi)/\sigma - f$. The r/σ term is:

(A) revenue net of the entry lottery cost

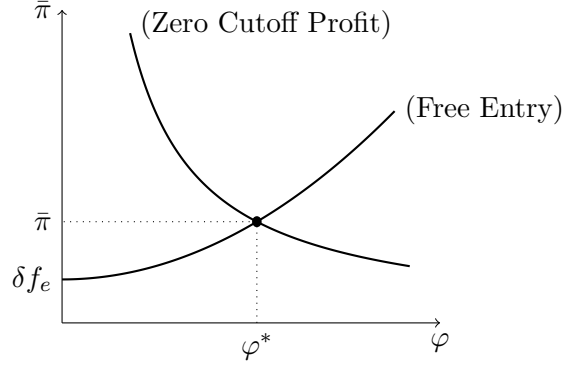


Figure 3: Closed-economy Melitz equilibrium. The downward-sloping ZCP curve and the upward-sloping FE curve (with intercept δf_e at $\varphi^* = 0$, where every entrant survives) intersect once, determining the cutoff φ^* and average profit $\bar{\pi}$.

- (B) variable profit, because the CES markup makes variable cost equal $\frac{\sigma-1}{\sigma}r$
- (C) the fixed cost share
- (D) the export premium

Q5.2 If firm A is twice as productive as firm B, its revenue is:

- (A) twice B's
- (B) $2^{\sigma-1}$ times B's
- (C) 2^σ times B's
- (D) the same, but at higher margin

Q5.3 In the $(\varphi^*, \bar{\pi})$ diagram, the equilibrium is where:

- (A) downward-sloping FE meets upward-sloping ZCP
- (B) downward-sloping ZCP meets upward-sloping FE
- (C) two upward-sloping curves cross
- (D) ZCP is vertical

6 Lesson 6: Opening the Melitz economy — selection and reallocation

Consider $n + 1$ symmetric countries. A firm may sell to each of the n foreign markets by paying a **per-market export fixed cost** $f_x > 0$ and iceberg cost $\tau > 1$.

Derivation: Who exports? The export cutoff

Per-market profits, conditional on selling there:

$$\pi_d(\varphi) = \frac{r_d(\varphi)}{\sigma} - f, \quad \pi_x(\varphi) = \frac{\tau^{1-\sigma} r_d(\varphi)}{\sigma} - f_x,$$

where $r_d(\varphi) = R\left(\frac{\sigma-1}{\sigma}\varphi P\right)^{\sigma-1}$ is domestic revenue and exports per market are $r_x(\varphi) = \tau^{1-\sigma} r_d(\varphi)$ (the iceberg cost raises the delivered price, shrinking revenue by $\tau^{1-\sigma}$). Both cutoffs come from zero profit:

$$r_d(\varphi^*) = \sigma f, \quad r_x(\varphi_x^*) = \sigma f_x.$$

Divide the second by the first, using the revenue ratio and $r_x = \tau^{1-\sigma} r_d$:

$$\frac{r_x(\varphi_x^*)}{r_d(\varphi^*)} = \tau^{1-\sigma} \left(\frac{\varphi_x^*}{\varphi^*} \right)^{\sigma-1} = \frac{f_x}{f} \quad \Rightarrow \quad \boxed{\varphi_x^* = \varphi^* \tau \left(\frac{f_x}{f} \right)^{\frac{1}{\sigma-1}}}$$

Provided $\tau^{\sigma-1} f_x > f$, we get $\varphi_x^* > \varphi^*$: **firms partition into three groups** — exit ($\varphi < \varphi^*$), serve only the home market ($\varphi^* \leq \varphi < \varphi_x^*$), and export to all markets ($\varphi \geq \varphi_x^*$). Only the most productive firms export: exactly the exporter premia in the data, generated by *self-selection*, not by exporting making firms better.

Derivation: Trade toughens selection: the cutoff rises

The open-economy ZCP gains an export term:

$$\bar{\pi} = f k(\varphi^*) + p_x n f_x k(\varphi_x^*), \quad p_x = \frac{1 - G(\varphi_x^*)}{1 - G(\varphi^*)},$$

where p_x is the probability a successful entrant exports, and φ_x^* is the function of φ^* derived above — so ZCP is still a curve in $(\varphi^*, \bar{\pi})$. The FE condition is *completely unchanged*: $\bar{\pi} = \delta f_e / [1 - G(\varphi^*)]$. Because the export term adds strictly positive expected profits at every φ^* , **ZCP shifts up**; FE stays put; the intersection moves up along FE:

$$\boxed{\varphi^* > \varphi_a^*}$$

— the survival cutoff is *tougher* under trade than in autarky (φ_a^*).

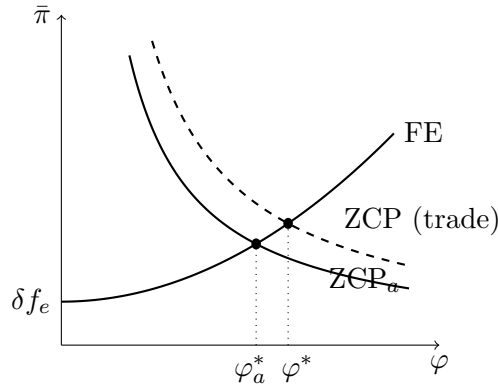


Figure 4: Impact of trade on selection. Export profits shift the ZCP curve up at every cutoff; the FE curve is unchanged. The equilibrium slides up along FE: the survival cutoff rises from φ_a^* (autarky) to φ^* (trade), forcing the least productive firms out.

Intuition: The reallocation mechanism, told through the labor market

Why should trade kill the weakest home-market firms, which do not even export? Follow the wage. Trade opens profitable foreign opportunities for the most productive firms; they expand, bidding up demand for labor. The real wage (relative to the price index) rises. Firms near the old cutoff, which broke even at the old factor costs, now make losses — they exit. Full employment then reallocates their workers to the expanding exporters. Sales of firms in $[\varphi_a^*, \varphi^*)$ go to zero (exit); purely domestic survivors *shrink* (their r_d falls); exporters may sell less at home but more in total (the export jump at φ_x^*). Market shares move up the productivity ladder.

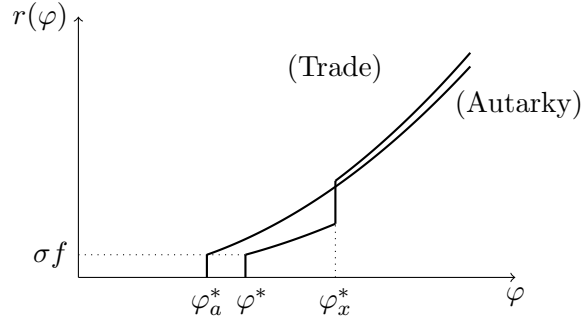


Figure 5: Impact of trade on firm revenues. Under trade the survival cutoff moves right ($\varphi^* > \varphi_a^*$; entrants below it earn zero). Purely domestic firms ($\varphi^* \leq \varphi < \varphi_x^*$) sell *less* than under autarky (their curve lies below); at φ_x^* revenue jumps discretely as firms add export markets, and beyond it exporters sell more than under autarky.

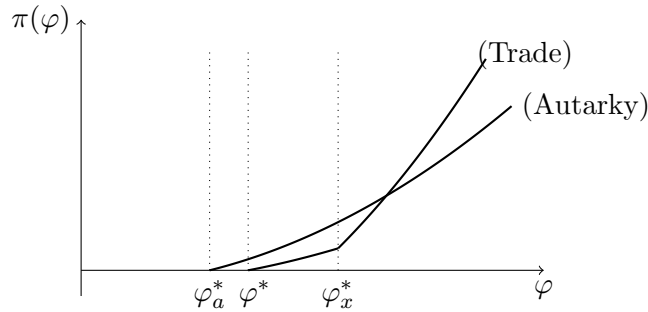


Figure 6: Impact of trade on firm profits. Profits are continuous but kinked at φ_x^* (the export fixed cost is exactly offset there). Firms in $[\varphi_a^*, \varphi^*)$ turn unprofitable and exit; mid-range survivors earn less than under autarky; sufficiently productive exporters earn more — the trade profit schedule crosses the autarky one from below.

Derivation: Welfare: aggregate productivity gains

Welfare per worker in any equilibrium can be reduced to the domestic cutoff. Using $W = M_t^{\frac{1}{1-\sigma}} \frac{\sigma-1}{\sigma} \tilde{\varphi}_t$ (with M_t varieties available and $\tilde{\varphi}_t$ their average productivity, both

cutoff-dependent) together with $r(\varphi^*) = \sigma f$ and the aggregation identities, one obtains the remarkably clean expressions

$$W_a = \frac{\sigma - 1}{\sigma} \left(\frac{L}{\sigma f} \right)^{\frac{1}{\sigma-1}} \varphi_a^*, \quad W = \frac{\sigma - 1}{\sigma} \left(\frac{L}{\sigma f} \right)^{\frac{1}{\sigma-1}} \varphi^*.$$

Welfare is *proportional to the survival cutoff*. Since trade raises the cutoff, $\varphi^* > \varphi_a^* \Rightarrow W > W_a$: **gains from trade via intra-industry reallocation** — a channel absent from both neoclassical models (no firms) and Krugman (identical firms).

Understanding Quiz 6 — Lesson 6

Q6.1 The condition $\tau^{\sigma-1} f_x > f$ guarantees that:

- (A) no firm exports
- (B) every survivor exports
- (C) the export cutoff exceeds the survival cutoff, so only some firms export
- (D) trade is welfare-reducing

Q6.2 When the economy opens to trade, in the $(\varphi^*, \bar{\pi})$ diagram:

- (A) FE shifts up, ZCP unchanged, cutoff falls
- (B) ZCP shifts up, FE unchanged, cutoff rises
- (C) both shift down
- (D) nothing moves; only prices change

Q6.3 The *new* channel of gains from trade in Melitz, relative to Krugman, is:

- (A) cheaper imported varieties
- (B) lower markups through competition
- (C) reallocation of labor and market share toward more productive firms
- (D) terms-of-trade improvements

Part III

Heterogeneous Firms and Distorted Gravity (Melitz–Pareto)

The original Melitz model is qualitative. To take it to data, assume productivity is **Pareto**. After a stretch of honest algebra, the model collapses into a structural gravity system as clean as EK's — but with a trade elasticity that mixes technology and taste parameters, and firm margins hidden inside.

7 Lesson 7: Setup — nested CES demand and Pareto firms

Definition: Environment

Countries $i, j = 1, \dots, N$; labor endowments L_i ; iceberg costs d_{ij} . Firms are heterogeneous in productivity φ . Fixed costs: prospective entrants in i pay $f_i^{(e)}$ (in domestic labor) to *draw* φ ; upon entry, selling to market j (including $j = i$) costs $f_{ij}^{(x)}$, paid in units of *destination* labor. Monopolistic competition.

Definition: Nested CES preferences

$$U_j = \left(\sum_i C_{ij}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad C_{ij} = \left(\int_{\omega \in \Omega_{ij}} c_{ij}(\omega)^{\frac{\gamma-1}{\gamma}} d\omega \right)^{\frac{\gamma}{\gamma-1}}.$$

Upper tier (elasticity $\sigma > 1$): substitution between country-of-origin composites — the Armington margin. *Lower tier* (elasticity $\gamma > 1$): substitution between individual firms within an origin — the Krugman/Melitz margin. Often $\gamma \geq \sigma$: two French firms' products are closer substitutes than “French goods” vs “German goods.” Firm demand chains the two tiers:

$$x_{ij}(\varphi) \equiv p_{ij}(\varphi)q_{ij}(\varphi) = \underbrace{\left(\frac{p_{ij}(\varphi)}{P_{ij}} \right)^{1-\gamma}}_{\text{firm's share within origin } i} \underbrace{\left(\frac{P_{ij}}{P_j} \right)^{1-\sigma}}_{\text{origin } i\text{'s sales } X_{ij}} X_j \quad (1)$$

with bilateral index $P_{ij} = \left[\int p_{ij}(\varphi)^{1-\gamma} M_i dG_i(\varphi) \right]^{\frac{1}{1-\gamma}}$ and consumer index $P_j = \left[\sum_i P_{ij}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$. Firm pricing carries the *lower-tier* markup:

$$p_{ij}(\varphi) = \bar{\gamma} \frac{d_{ij} w_i}{\varphi}, \quad \bar{\gamma} \equiv \frac{\gamma}{\gamma-1}. \quad (2)$$

Definition: Pareto productivity

$$G_i(\varphi) = 1 - \left(\frac{\varphi}{a_i} \right)^{-\theta}, \quad g_i(\varphi) = \theta a_i^\theta \varphi^{-\theta-1}, \quad (5)$$

with shape $\theta > 1$ (dispersion: higher θ = more similar firms) and country scale a_i . The crucial property (proved in the EK companion): **a Pareto truncated from below is Pareto with the same θ** . Since Melitz selection is exactly truncation from below at φ_{ij}^* , the distribution of *active* firms stays Pareto — this is why everything will integrate into clean power functions.

Intuition: What each parameter will do

Keep a scorecard: γ governs competition *between firms* (markups, firm-level demand curves); σ governs substitution *between countries*; θ governs how heterogeneous firms are, hence how strong selection margins are. The punchline of this part is which combination of the three ends up as the *aggregate* trade elasticity — and it will not be the one Armington intuition suggests.

Understanding Quiz 7 — Lesson 7

Q7.1 In the nested CES structure, the elasticity γ applies to substitution:

- (A) between origin-country composites
- (B) between individual firms/products within an origin
- (C) between labor and capital
- (D) between traded and nontraded goods

Q7.2 Firms price at markup $\bar{\gamma} = \gamma/(\gamma - 1)$ (not $\sigma/(\sigma - 1)$) because:

- (A) each firm is atomistic and competes with other firms in the lower tier, whose elasticity is γ
- (B) the upper tier does not exist for exporters
- (C) regulation caps markups
- (D) γ is always smaller than σ

Q7.3 Selection (only firms above φ_{ij}^* serve j) keeps the active-firm distribution tractable because:

- (A) truncated Pareto is still Pareto with the same shape θ
- (B) Pareto and Fréchet are the same distribution
- (C) the cutoff is always at the mean
- (D) G_i becomes uniform after truncation

8 Lesson 8: Aggregating firms — trade flows, price indices, and the cutoff

8.1 Bilateral trade flows

Derivation: Integrating firm exports over the Pareto tail

Total flow from i to j sums firm-level sales over active firms ($\varphi \geq \varphi_{ij}^*$):

$$X_{ij} = M_i \int_{\varphi_{ij}^*}^{\infty} x_{ij}(\varphi) dG_i(\varphi) = M_i \int_{\varphi_{ij}^*}^{\infty} \left(\frac{p_{ij}(\varphi)}{P_j} \right)^{1-\gamma} \left(\frac{P_{ij}}{P_j} \right)^{1-\sigma} X_j dG_i(\varphi).$$

Pull out everything not involving φ , substitute $p_{ij}(\varphi) = \bar{\gamma} d_{ij} w_i / \varphi$ and $dG_i = \theta a_i^\theta \varphi^{-\theta-1} d\varphi$:

$$X_{ij} = M_i P_{ij}^{\gamma-1} \left(\frac{P_{ij}}{P_j} \right)^{1-\sigma} X_j (\bar{\gamma} d_{ij} w_i)^{1-\gamma} a_i^\theta \theta \int_{\varphi_{ij}^*}^{\infty} \varphi^{\gamma-\theta-2} d\varphi.$$

(The φ -powers: $(1/\varphi)^{1-\gamma} = \varphi^{\gamma-1}$ times $\varphi^{-\theta-1}$ gives $\varphi^{\gamma-\theta-2}$.) The tail integral converges iff $\gamma - \theta - 1 < 0$, i.e.

$$\boxed{\theta > \gamma - 1} \quad \text{and then} \quad \int_{\varphi_{ij}^*}^{\infty} \varphi^{\gamma-\theta-2} d\varphi = \frac{(\varphi_{ij}^*)^{\gamma-\theta-1}}{\theta - \gamma + 1}.$$

Hence

$$X_{ij} = M_i P_{ij}^{\gamma-1} \left(\frac{P_{ij}}{P_j} \right)^{1-\sigma} X_j (\bar{\gamma} d_{ij} w_i)^{1-\gamma} a_i^\theta \frac{\theta}{\theta - \gamma + 1} (\varphi_{ij}^*)^{\gamma-\theta-1}. \quad (6)$$

Intuition: Why $\theta > \gamma - 1$?

Sales rise with productivity like $\varphi^{\gamma-1}$; the Pareto density thins like $\varphi^{-\theta-1}$. If firms are too heterogeneous (θ small) relative to how strongly sales reward productivity (γ big), a handful of superstars would carry infinite sales. $\theta > \gamma - 1$ keeps the superstar tail integrable — the same tension as $\sigma < 1 + \theta$ in EK.

Derivation: Bilateral price index

Identical mechanics on $P_{ij}^{1-\gamma} = M_i \int_{\varphi_{ij}^*}^{\infty} (\bar{\gamma} d_{ij} w_i / \varphi)^{1-\gamma} dG_i(\varphi)$:

$$P_{ij}^{1-\gamma} = \frac{\theta}{\theta - \gamma + 1} a_i^\theta M_i (\bar{\gamma} d_{ij} w_i)^{1-\gamma} (\varphi_{ij}^*)^{\gamma-\theta-1}. \quad (7)$$

Note the composition effect: a *higher cutoff* φ_{ij}^* removes the worst (most expensive) suppliers, and since $\gamma - \theta - 1 < 0$ it *lowers* $P_{ij}^{1-\gamma}$'s bracket appropriately — selection shows up directly in prices.

8.2 The zero-cutoff-profit condition

Derivation: Which firms serve market j ?

With CES, variable profit is a $1/\gamma$ share of sales (lower-tier markup), so profits of firm φ from market j :

$$\pi_{ij}(\varphi) = \frac{1}{\gamma} p_{ij}(\varphi) q_{ij}(\varphi) - w_j f_{ij}^{(x)}, \quad p_{ij}(\varphi) q_{ij}(\varphi) = \left(\bar{\gamma} \frac{d_{ij} w_i}{\varphi} \right)^{1-\gamma} P_{ij}^{\gamma-1} X_{ij}.$$

The marginal entrant to market j breaks even, $\pi_{ij}(\varphi_{ij}^*) = 0$:

$$w_j f_{ij}^{(x)} = \frac{1}{\gamma} \left(\bar{\gamma} \frac{d_{ij} w_i}{\varphi_{ij}^*} \right)^{1-\gamma} P_{ij}^{\gamma-1} X_{ij} \implies \varphi_{ij}^* = \left(\gamma w_j f_{ij}^{(x)} \right)^{-\frac{1}{1-\gamma}} X_{ij}^{\frac{1}{1-\gamma}} \frac{\bar{\gamma} d_{ij} w_i}{P_{ij}}. \quad (8)$$

Read it economically: the cutoff is *higher* (tougher selection into j) when the market-entry fee $w_j f_{ij}^{(x)}$ is larger, when delivered costs $d_{ij} w_i$ are higher, and when competition in the i -to- j segment is fiercer (P_{ij} low); it is *lower* when the destination is big (X_{ij} large — note $\frac{1}{1-\gamma} < 0$ flips signs appropriately).

Understanding Quiz 8 — Lesson 8

Q8.1 The convergence condition $\theta > \gamma - 1$ is needed because otherwise:

- (A) no firm would enter
- (B) the sales of the most productive (fat-tail) firms would make aggregate sales infinite
- (C) the price index would be negative
- (D) the cutoff would be below a_i

Q8.2 In the ZCP equation, a larger destination market (X_{ij} or L_j up) makes the entry cutoff

φ_{ij}^* :

- (A) lower — bigger markets make the fixed cost easier to cover, so weaker firms enter
- (B) higher — bigger markets are more competitive
- (C) unchanged — cutoffs depend only on $f^{(x)}$
- (D) undefined

Q8.3 Variable profits are the fraction $1/\gamma$ of sales because:

- (A) fixed costs are $1/\gamma$ of revenue
- (B) with markup $\gamma/(\gamma - 1)$, variable costs absorb $(\gamma - 1)/\gamma$ of revenue
- (C) the Pareto shape imposes it
- (D) destination labor is paid $1/\gamma$ of sales

9 Lesson 9: Distorted gravity — marketing costs, free entry, and the trade elasticity

9.1 A hidden constant: marketing costs as a share of sales

Derivation: Aggregate marketing (market-access) costs

Total fixed market-access spending by i -firms selling in j : mass of participants $\times$ fee. Participation probability is the Pareto tail $\Pr(\varphi \geq \varphi_{ij}^*) = (\varphi_{ij}^*/a_i)^{-\theta}$:

$$\Gamma_{ij} = M_i w_j f_{ij}^{(x)} \left(\frac{\varphi_{ij}^*}{a_i} \right)^{-\theta}.$$

Now combine the price index (7) with the cutoff (8). Substituting (7) into (8) and reorganizing (a few lines of exponent bookkeeping whose engine is, again, that all objects are powers of φ_{ij}^*) yields the identity

$$M_i w_j f_{ij}^{(x)} \left(\frac{\varphi_{ij}^*}{a_i} \right)^{-\theta} = \frac{\theta - \gamma + 1}{\theta} \cdot \frac{X_{ij}}{\gamma} \quad \Rightarrow \quad \boxed{\Gamma_{ij} = \frac{\theta - \gamma + 1}{\theta \gamma} X_{ij}} \quad (9)$$

Aggregate fixed market-access costs are a *constant share* of bilateral sales — the same share for every country pair, pinned by θ and γ only. This is the Pareto magic: cutoffs, masses, and fees all conspire into one number.

Derivation: Free entry pins down the mass of entrants

Aggregate profits of country i 's firms: variable profits (X_{ij}/γ per market) minus marketing costs:

$$\Pi_i = \sum_j \left[\frac{1}{\gamma} X_{ij} - \Gamma_{ij} \right] = \sum_j \left(\frac{1}{\gamma} - \frac{\theta - \gamma + 1}{\theta \gamma} \right) X_{ij} = \frac{\gamma - 1}{\theta \gamma} Y_i,$$

where $Y_i = \sum_j X_{ij} = w_i L_i$ is total sales. Free entry equates expected profits to total entry spending $w_i f_i^{(e)} M_i$:

$$w_i f_i^{(e)} M_i = \frac{\gamma - 1}{\theta \gamma} Y_i \quad \Rightarrow \quad \boxed{M_i = \frac{\gamma - 1}{\theta \gamma} \frac{L_i}{f_i^{(e)}}} \quad (10)$$

As in Krugman: the mass of entrants is proportional to population — but now θ tempers it, because part of profits is eaten by market-access fees.

9.2 Structural gravity

Derivation: Closing the system

Two equations in two unknowns per pair, given wages: the ZCP (8) gives φ_{ij}^* as a function of P_{ij} (and P_j , L_j); the price index (7), with M_i from (10), gives P_{ij} as a function of φ_{ij}^* . Solving the pair (substitute one into the other and collect the exponents of P_{ij}) delivers a pure power function:

$$P_{ij}^{1-\sigma} = \Lambda_{ij} (d_{ij} w_i)^{-\theta \lambda} P_j^{-\frac{(1-\sigma)}{1-\gamma} (\gamma - \theta - 1) \lambda}, \quad \lambda \equiv \left[1 + \theta \left(\frac{1}{\sigma - 1} - \frac{1}{\gamma - 1} \right) \right]^{-1},$$

where Λ_{ij} bundles primitives:

$$\Lambda_{ij} \equiv \left(\frac{1}{\theta - \gamma + 1} \frac{\gamma - 1}{\gamma} a_i^\theta \frac{L_i}{f_i^{(e)}} \bar{\gamma}^{-\theta} \right)^\lambda \left(\frac{L_j}{\gamma f_{ij}^{(x)}} \right)^{\lambda \frac{\gamma - \theta - 1}{1 - \gamma}}. \quad (15)$$

Summing $P_{ij}^{1-\sigma}$ over origins and solving for P_j :

$$P_j = \left(\sum_\ell \Lambda_{\ell j} (d_{\ell j} w_\ell)^{-\theta \lambda} \right)^{-\frac{1}{\theta \lambda}}, \quad \boxed{\pi_{ij} \equiv \frac{X_{ij}}{X_j} = \left(\frac{P_{ij}}{P_j} \right)^{1-\sigma} = \frac{\Lambda_{ij} (d_{ij} w_i)^{-\theta \lambda}}{\sum_\ell \Lambda_{\ell j} (d_{\ell j} w_\ell)^{-\theta \lambda}}} \quad (12-13)$$

Structural gravity again — with **trade elasticity** $\theta \lambda$. General equilibrium: wages such that (12)–(13) hold and $w_i L_i = \sum_j \pi_{ij} w_j L_j$.

Intuition: Why “distorted” gravity — and what the elasticity means

The model looks exactly like Armington/EK gravity from the outside, but the elasticity is the composite

$$\theta \lambda = \theta \left[1 + \theta \left(\frac{1}{\sigma - 1} - \frac{1}{\gamma - 1} \right) \right]^{-1}.$$

Special case $\sigma = \gamma$: $\lambda = 1$ and the trade elasticity is just θ — the Pareto shape. *The substitution elasticities drop out of the aggregate trade elasticity entirely!* The reason is

Chaney's (2008) margin decomposition:

$$\varepsilon \equiv -\frac{\partial \ln X_{ij}}{\partial \ln d_{ij}} = \underbrace{(\sigma - 1)}_{\text{intensive margin}} + \underbrace{\theta - (\sigma - 1)}_{\text{extensive margin}} = \theta.$$

When trade costs fall, existing exporters sell more (intensive, strength $\sigma - 1$) *and* new, less productive firms start exporting (extensive). With Pareto firms the two margins are perfect complements: if goods are highly substitutable (σ big), the intensive response is big, but the new marginal exporters are tiny, so the extensive response is small — the total is always θ . Elasticity of trade = technology dispersion, exactly as in EK, though the micro-mechanism is completely different (entry and selection rather than min-cost sourcing).

Derivation: Welfare: same sufficient statistic, different elasticity

In “hat” notation (counterfactual changes), welfare changes from trade-cost shocks satisfy

$$\hat{U}_j = \frac{\hat{w}_j}{\hat{P}_j} = (\hat{\pi}_{jj})^{-\frac{1}{\theta\lambda}} \quad \text{vs Armington:} \quad \hat{U}_j = (\hat{\pi}_{jj})^{-\frac{1}{\sigma-1}}.$$

The domestic share is a sufficient statistic in both (the ACR result); models differ only through the elasticity that scales it. All the Melitz machinery — cutoffs, selection, firm entry — compresses into $\theta\lambda$.

Understanding Quiz 9 — Lesson 9

Q9.1 In Melitz–Pareto, aggregate fixed market-access costs Γ_{ij} equal:

- (A) a constant share $\frac{\theta-\gamma+1}{\theta\gamma}$ of bilateral sales X_{ij}
- (B) exactly the wage bill of country j
- (C) zero in equilibrium
- (D) a share that grows with distance

Q9.2 When $\sigma = \gamma$, the aggregate trade elasticity equals:

- (A) $\sigma - 1$
- (B) $\gamma - 1$
- (C) θ
- (D) $\theta(\sigma - 1)$

Q9.3 In Chaney's decomposition, a fall in d_{ij} raises trade through:

- (A) only incumbent exporters selling more
- (B) only new firms starting to export
- (C) both: incumbents sell more (intensive, $\sigma - 1$) and new firms enter (extensive, $\theta - (\sigma - 1)$)
- (D) neither; only prices change

Final Exam: Comprehensive Multiple-Choice Quiz

Understanding Quiz — Final Exam (all parts)

F1. The fundamental driver of trade in Krugman (1979, 1980), absent from neoclassical models, is:

- (A) comparative advantage
- (B) factor endowment differences
- (C) increasing returns to scale with love of variety
- (D) exchange-rate policy

F2. In Krugman-1979, opening two identical economies to trade lowers p/w because:

- (A) wages fall
- (B) per-capita consumption of each variety falls, demand becomes more elastic, and markups shrink
- (C) productivity rises exogenously
- (D) fixed costs disappear

F3. Which object is *endogenous* in both Krugman models but exogenous in Armington?

- (A) the wage
- (B) the set/mass of varieties
- (C) the trade cost
- (D) the labor force

F4. In Krugman-1980 the equilibrium firm scale is $q_i = (\sigma - 1)\varphi_i f_i$. Higher fixed costs f_i imply:

- (A) smaller firms and more of them
- (B) larger firms and fewer of them
- (C) larger firms and more of them
- (D) no change in scale

F5. The Krugman-1980 real-income equation $w_i/P_i = A_i L_i^{\frac{1}{\sigma-1}} \pi_{ii}^{-\frac{1}{\sigma-1}}$ implies that, holding openness fixed, population raises welfare because:

- (A) wages are higher in big countries by assumption
- (B) more people sustain more varieties, and consumers love variety
- (C) bigger countries have lower trade costs
- (D) the markup falls with population

F6. Empirically, exporters are:

- (A) a majority of firms, slightly larger than average
- (B) a small minority, and larger, more productive, more capital- and skill-intensive
- (C) randomly drawn from the firm distribution
- (D) concentrated among the least productive firms

F7. In Melitz, the zero-cutoff-profit and free-entry conditions jointly determine:

- (A) $(\varphi^*, \bar{\pi})$: the survival cutoff and average profit
- (B) the wage and the exchange rate
- (C) σ and θ
- (D) the iceberg cost

F8. The Melitz export cutoff satisfies $\varphi_x^* = \varphi^* \tau (f_x/f)^{\frac{1}{\sigma-1}}$. Exporting is rarer when:

- (A) τ or f_x is higher
- (B) f is higher
- (C) $\sigma \rightarrow \infty$ with $f_x < f$
- (D) markets are larger

F9. Opening a Melitz economy to trade raises welfare *even for consumers of purely domestic goods* because:

- (A) domestic firms cut markups
- (B) the survival cutoff rises: low-productivity firms exit and resources shift to high-productivity firms, raising aggregate productivity
- (C) the government redistributes export profits
- (D) trade lowers fixed costs

F10. In Melitz–Pareto, aggregate variable profits are X_{ij}/γ and market-access costs are $\frac{\theta-\gamma+1}{\theta\gamma}X_{ij}$, so aggregate net profits are the fraction of sales:

- (A) $\frac{1}{\theta}$
- (B) $\frac{\gamma-1}{\theta\gamma}$
- (C) $\frac{1}{\gamma}$
- (D) $\frac{\theta-1}{\gamma}$

F11. The Melitz–Pareto trade elasticity is $\theta\lambda$ with $\lambda = [1 + \theta(\frac{1}{\sigma-1} - \frac{1}{\gamma-1})]^{-1}$. When $\sigma = \gamma$:

- (A) the elasticity is $\sigma - 1$, as in Armington
- (B) the elasticity is θ , and substitution elasticities play no role in it
- (C) the elasticity is infinite
- (D) trade collapses

F12. Rank the models by the *interpretation* of their trade elasticity: Armington/Krugman vs EK/Melitz–Pareto.

- (A) taste ($\sigma - 1$) in both groups
- (B) technology (θ) in both groups
- (C) taste ($\sigma - 1$) in Armington/Krugman; technology-heterogeneity (θ) in EK/Melitz–Pareto
- (D) taste in EK; technology in Krugman

F13. The welfare formula $\hat{U}_j = \hat{\pi}_{jj}^{-1/\epsilon}$ (with ϵ the trade elasticity) means that two models matched to the same data on domestic shares:

- (A) always deliver identical gains from trade
- (B) deliver the same gains iff they share the same trade elasticity, whatever their micro-structure
- (C) never deliver the same gains
- (D) cannot be compared

F14. When trade costs fall in Melitz–Pareto and σ is very large, the total trade response is still θ (when $\sigma = \gamma$) because:

- (A) the intensive margin dominates and equals θ
- (B) a strong intensive response is exactly offset by a weak extensive response: new marginal exporters are tiny when goods are close substitutes
- (C) the extensive margin is always zero
- (D) fixed costs adjust endogenously

A Answer key

Lesson quizzes

Lesson 1

- Q1.1:** B. $\partial\varepsilon/\partial c < 0$ by assumption; $\mu = \varepsilon/(\varepsilon - 1)$ falls in ε , so lower ε means higher markup.
Q1.2: B. ZZ is price = average cost; average cost $w(f/q + 1/\varphi)$ falls with output $q = Lc$.
Q1.3: B. $M = L/(\varepsilon f)$: doubling f halves the varieties the labor force can sustain.

Lesson 2

- Q2.1:** C. PP has no market-size term; ZZ's $f/(qL)$ -term shrinks when the market doubles.
Q2.2: B. From free entry, $q = f/(p/w - 1/\varphi)$; a smaller markup wedge requires bigger scale to cover f . **Q2.3:** C. $M_1 < M_0$ (labor constraint with bigger firms) but $2M_1 > M_0$ (shown via $c_1 < c_0$).

Lesson 3

- Q3.1:** B. CES $\Rightarrow$ constant $\varepsilon = \sigma \Rightarrow$ constant markup $\sigma/(\sigma - 1)$. **Q3.2:** B. $\pi_{ij} \propto d_{ij}^{1-\sigma}$: the elasticity is $1 - \sigma$, reflecting substitution between varieties. **Q3.3:** B. Scale economies favor concentrating production; trade costs favor locating near demand; hence big home markets become export bases.

Lesson 4

- Q4.1:** A. About 4% economy-wide (18% within manufacturing). **Q4.2:** B. Extreme concentration of export value in few firms is the signature of a fat-tailed distribution. **Q4.3:** D. Exporters pay *higher* wages (premium ≈ 0.17 log points).

Lesson 5

- Q5.1:** B. With markup $\sigma/(\sigma - 1)$, variable cost is $\frac{\sigma-1}{\sigma}r$, leaving r/σ as variable profit.
Q5.2: B. $r(\varphi_1)/r(\varphi_2) = (\varphi_1/\varphi_2)^{\sigma-1}$. **Q5.3:** B. ZCP decreasing (via $k(\varphi^*)$), FE increasing (rarer success requires higher $\bar{\pi}$); unique crossing.

Lesson 6

- Q6.1:** C. It makes $\varphi_x^* > \varphi^*$, generating the three-way partition (exit / domestic only / exporters). **Q6.2:** B. Export profits add to expected profits at any cutoff (ZCP up); the entry lottery is unchanged (FE fixed); so φ^* rises. **Q6.3:** C. Intra-industry reallocation toward high- φ firms raises aggregate productivity — the Melitz channel.

Lesson 7

- Q7.1:** B. γ is the lower-tier (within-origin, firm-level) elasticity; σ is across origins. **Q7.2:** A. Each atomistic firm's residual demand slope is governed by within-nest substitution γ .
Q7.3: A. Truncation from below preserves the Pareto shape θ (only the scale moves), so selection never breaks the functional form.

Lesson 8

Q8.1: B. The integrand $\varphi^{\gamma-\theta-2}$ must vanish fast enough at ∞ ; otherwise superstar firms carry unbounded sales. **Q8.2: A.** From (8), $\varphi_{ij}^* \propto X_{ij}^{\frac{1}{1-\gamma}}$ with $\frac{1}{1-\gamma} < 0$: bigger markets admit weaker firms. **Q8.3: B.** Price $= \frac{\gamma}{\gamma-1} \times$ marginal cost $\Rightarrow$ cost share of revenue is $\frac{\gamma-1}{\gamma}$.

Lesson 9

Q9.1: A. Equation (9): the Pareto structure makes access costs a universal constant share of sales. **Q9.2: C.** $\lambda = 1$, so the elasticity collapses to the Pareto shape θ . **Q9.3: C.** Both margins operate; their sum is θ under $\sigma = \gamma$.

Final exam

Final Exam key

F1: C. Identical countries trade because scale economies + love of variety create gains even without comparative advantage.

F2: B. The pro-competitive effect: bigger market $\rightarrow$ lower c per variety $\rightarrow$ higher $\varepsilon \rightarrow$ lower markup.

F3: B. Free entry makes the variety set endogenous; Armington endows each country with one fixed variety.

F4: B. q_i scales with f_i (need more revenue to cover fixed costs); $M_i = L_i/(\sigma f_i)$ falls.

F5: B. Pure variety-scale effect, elasticity $1/(\sigma - 1)$.

F6: B. The exporter-premia facts (BJRS 2007).

F7: A. ZCP (down) and FE (up) intersect at $(\varphi^*, \bar{\pi})$; everything else follows by accounting.

F8: A. Higher variable or fixed export costs push φ_x^* up, shrinking the set of exporters.

F9: B. The reallocation/selection channel: welfare is proportional to φ^* , which rises with openness.

F10: B. $\frac{1}{\gamma} - \frac{\theta-\gamma+1}{\theta\gamma} = \frac{\theta-\theta+\gamma-1}{\theta\gamma} = \frac{\gamma-1}{\theta\gamma}$.

F11: B. $\lambda = 1 \Rightarrow$ elasticity $= \theta$; taste parameters vanish from the aggregate elasticity.

F12: C. Armington/Krugman: $\sigma - 1$ (differentiation); EK/Melitz-Pareto: θ (productivity heterogeneity). Same gravity form, different economics.

F13: B. ACR sufficient-statistic result: given data $(\hat{\pi}_{jj})$ and a common elasticity, micro-structure does not change measured gains.

F14: B. High σ strengthens the intensive margin $(\sigma - 1)$ but the marginal entrants are negligible sellers, weakening the extensive margin $(\theta - (\sigma - 1))$; with Pareto the sum is invariant.